西北农林科技大学

Northwest A&F University

流式计算实验报告

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环境准备

```
-bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
root@master:~# /etc/init.d/mysql start
 * Starting MySQL database server mysqld
su: warning: cannot change directory to /nonexistent: No such file or directory
                                                                        [ OK ]
root@master:~# start-cluster.sh
Starting cluster.
Starting standalonesession daemon on host master.
Please specify JAVA_HOME. Either in Flink config ./conf/flink-conf.yaml or as sy
stem-wide JAVA_HOME.
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stem-wide JAVA_HOME.
root@master:~#
```

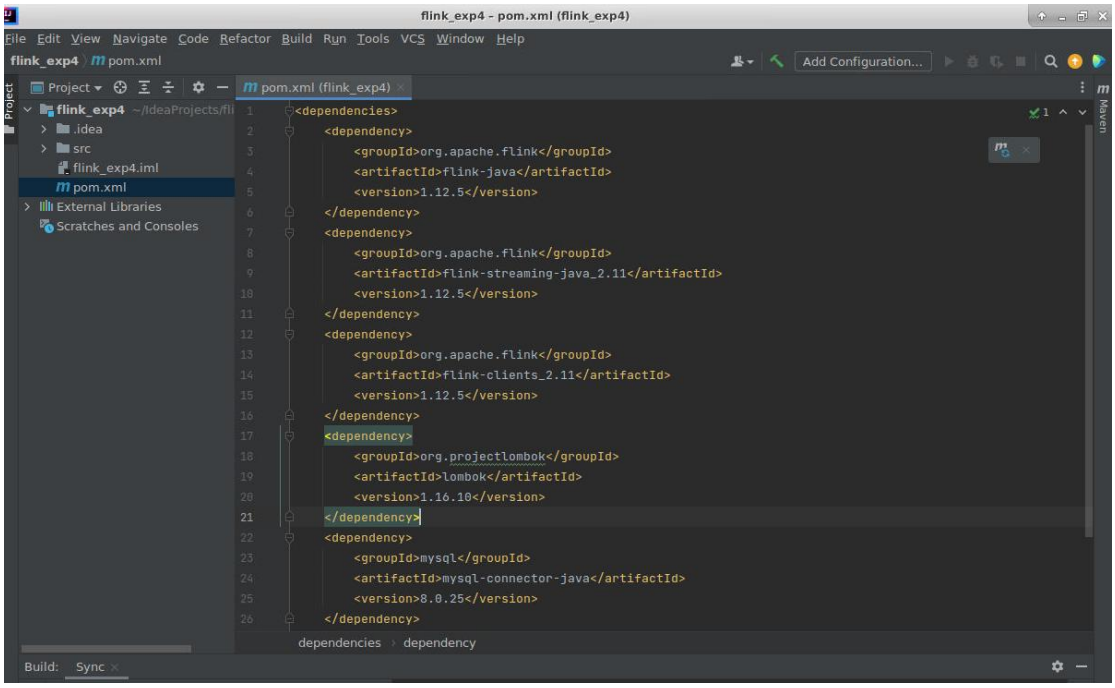

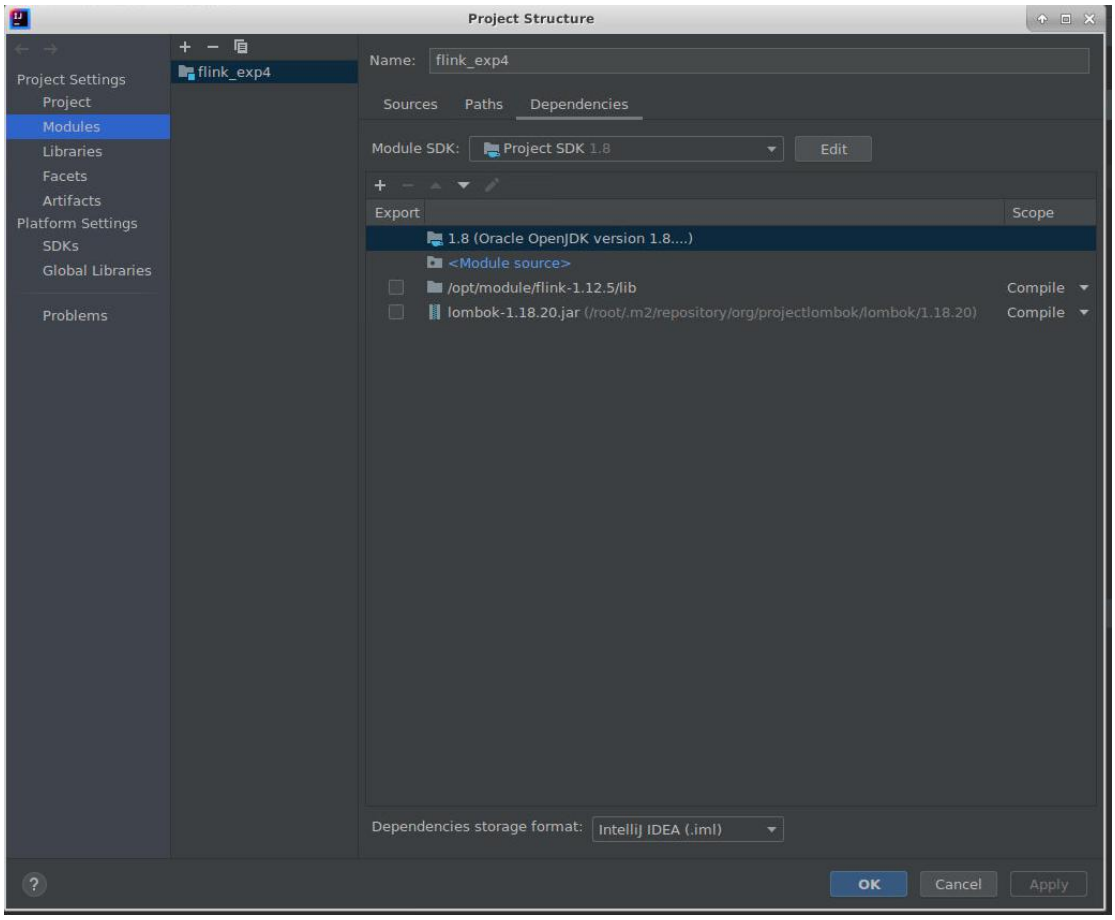

将 DataStream 注册为 Table 和 View 并进行 SQL 统计

```
root
 |-- user: BIGINT
 |-- product: STRING
 |-- amount: INT

86> (true,FlinkSQL_Table_Demo0.Order(user=2, product=pen, amount=3))
92> (true,FlinkSQL_Table_Demo0.Order(user=1, product=diaper, amount=4))
87> (true,FlinkSQL_Table_Demo0.Order(user=1, product=beer, amount=3))
87> (true,FlinkSQL_Table_Demo0.Order(user=2, product=rubber, amount=3))
```

使用 SQL 和 Table 两种方式对 DataStream 中的单词进行统计

SQL

```
23> (true,FlinkSQL_Table_Demo1.WC(word=World, frequency=1))
40> (true,FlinkSQL_Table_Demo1.WC(word=Hello, frequency=1))
40> (false,FlinkSQL_Table_Demo1.WC(word=Hello, frequency=1))
40> (true,FlinkSQL_Table_Demo1.WC(word=Hello, frequency=2))
```

Table

```
ERROR StatusLogger No Log4j 2 configuration file found. Using de
40> (true,FlinkSQL_Table_Demo2.WC(word=Hello, frequency=2))
```

使用 Flink SQL 统计用户订单信息

```
ERROR StatusLogger No Log4j 2 configuration file found. Using default co
87> (true,0,74,3)
35> (true,2,14,12)
36> (true,1,46,46)
87> (true,0,15,4)
36> (true,1,71,13)
35> (true,2,57,57)
87> (true,0,18,18)
36> (true,1,14,14)
35> (true,2,77,12)
87> (true,0,14,14)
```